



D.1 Gravitational fields

Fields · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Newton's law of gravitation states that any two point masses attract each other with a force $F = Gm_1m_2/r^2$, where G is the universal gravitational constant and r is the separation between centres; this produces a gravitational field of strength $g = GM/r^2$ around a mass M , directed towards it, equal to the gravitational force per unit mass on a small test mass. Gravitational potential V_g at a point is the work done per unit mass to bring a small test mass from infinity to that point, $V_g = -GM/r$ (taking $V_g = 0$ at infinity), and gravitational potential energy of a mass m is $E_p = mV_g = -GMm/r$. For a satellite in a stable circular orbit, gravitational force provides the centripetal force: $GMm/r^2 = mv^2/r = m\omega^2r$, giving orbital speed $v = \sqrt{GM/r}$ and period $T = 2\pi\sqrt{r^3/GM}$ (Kepler's third law).

Worked example: follow the reasoning

A satellite orbits Earth ($M = 5.97 \times 10^{24}$ kg) at radius $r = 7.0 \times 10^6$ m. Find its orbital speed and period. ($G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\cdot\text{kg}^{-2}$)
 $v = \sqrt{GM/r} = \sqrt{[(6.67 \times 10^{-11})(5.97 \times 10^{24})]/(7.0 \times 10^6)} = \sqrt{[3.982 \times 10^{14}/7.0 \times 10^6]} = \sqrt{(5.689 \times 10^7)} = 7543 \text{ m}\cdot\text{s}^{-1}$.
 $T = 2\pi r/v = 2\pi(7.0 \times 10^6)/7543 = 4.398 \times 10^7/7543 = 5830 \text{ s}$.
 Answer: $v \approx 7.5 \times 10^3 \text{ m}\cdot\text{s}^{-1}$, $T \approx 5.8 \times 10^3 \text{ s}$ ($\approx 97 \text{ min}$).

Quick check · multiple choice

What happens to the gravitational field strength at the surface of a planet if the planet's radius is doubled while its mass stays the same?

- A Doubles
- B Halves
- C Quarters
- D Stays the same

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A planet has mass 4.0×10^{24} kg and radius 5.0×10^6 m. ($G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2\cdot\text{kg}^{-2}$) [8]
 - (a) Calculate the gravitational field strength at the planet's surface.
 - (b) Calculate the gravitational potential at the surface.
 - (c) Calculate the minimum kinetic energy (escape energy) a 500 kg probe would need at the surface to escape the planet's gravitational field completely, and hence its escape speed.

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $g = GM/r^2 = (6.67 \times 10^{-11})(4.0 \times 10^{24})/(5.0 \times 10^6)^2 = (2.668 \times 10^{14})/(2.5 \times 10^{13}) = 10.7 \text{ m}\cdot\text{s}^{-2}$.

(b) $V_g = -GM/r = -(6.67 \times 10^{-11})(4.0 \times 10^{24})/(5.0 \times 10^6) = -2.668 \times 10^{14}/5.0 \times 10^6 = -5.34 \times 10^7 \text{ J}\cdot\text{kg}^{-1}$.

(c) To escape, total energy ≥ 0 : $\frac{1}{2}mv_{\text{esc}}^2 + mV_g = 0 \rightarrow \frac{1}{2}v_{\text{esc}}^2 = -V_g = 5.34 \times 10^7 \rightarrow v_{\text{esc}}^2 = 1.068 \times 10^8 \rightarrow v_{\text{esc}} = 1.03 \times 10^4 \text{ m}\cdot\text{s}^{-1}$. Escape KE = $\frac{1}{2}mv_{\text{esc}}^2 = \frac{1}{2}(500)(1.068 \times 10^8) = 2.67 \times 10^{10} \text{ J}$.

Quick check answer

C. $g = GM/r^2$; doubling r with M constant reduces g by a factor of $2^2 = 4$, so g becomes one quarter of its original value.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.